SIMATS ENGINEERING

C03 ASSESSMENT TOOL1

COURSE CODE: CSA1610 COURSE NAME: DWDM

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EXPERIMENTS

Q1 R Code for Covariance and Correlation:

CODE:

@relation photo_preference

@attribute Age_of_child {5-6,7-8,9-10}

@attribute Photo_A numeric

@attribute Photo_B numeric

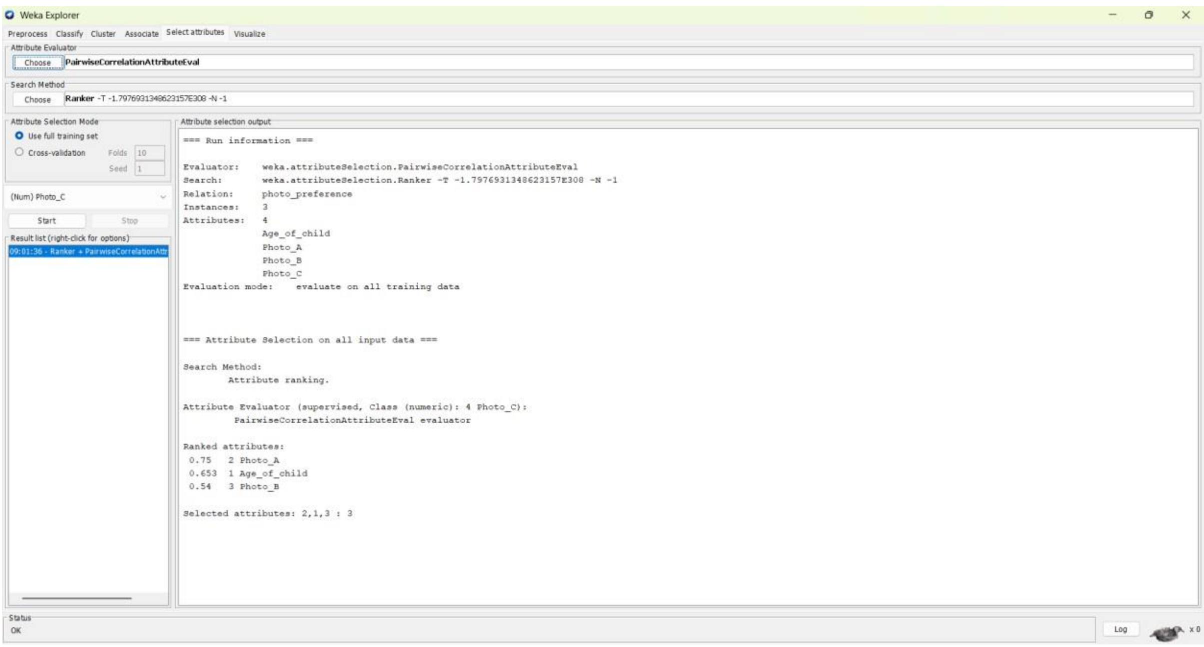
@attribute Photo_C numeric

@data

5-6,18,22,20

7-8,2,28,40

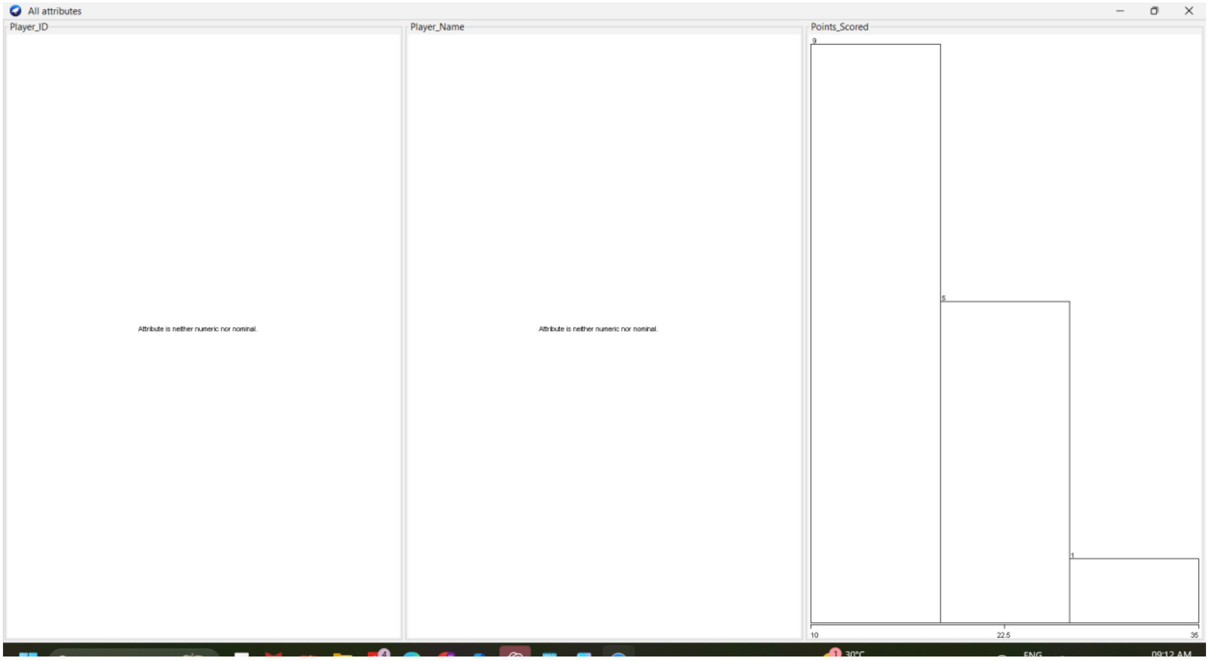
9-10,20,10,40



Q2 R Code for Boxplot and Outlier Detection

CODE:

```
@relation tennis_scores

@attribute Player_ID string
@attribute Player_Name string
@attribute Points_Scored numeric

@data
P1,Player_A,12
P2,Player_B,15
P3,Player_C,14
P4,Player_D,10
P5,Player_E,18
P6,Player_F,20
P7,Player_G,22
P8,Player_H,13
P9,Player_I,11
P10,Player_J,35
P11,Player_K,16
P12,Player_L,17
P13,Player_M,19
P14,Player_N,21
P15,Player_O,23
```



Q3 R Code for Decision Tree Classifier

CODE:

```
@relation loan_approval


@attribute Age {young,middle,old}
@attribute Income {low,medium,high}
@attribute Employment_Status {employed,unemployed,self-employed}
@attribute Credit_Score {poor,average,good}
@attribute Loan_Approval {yes,no}


@data
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
```

young,medium,employed,average,yes

young,low,unemployed,poor,no

old,medium,self-employed,good,yes

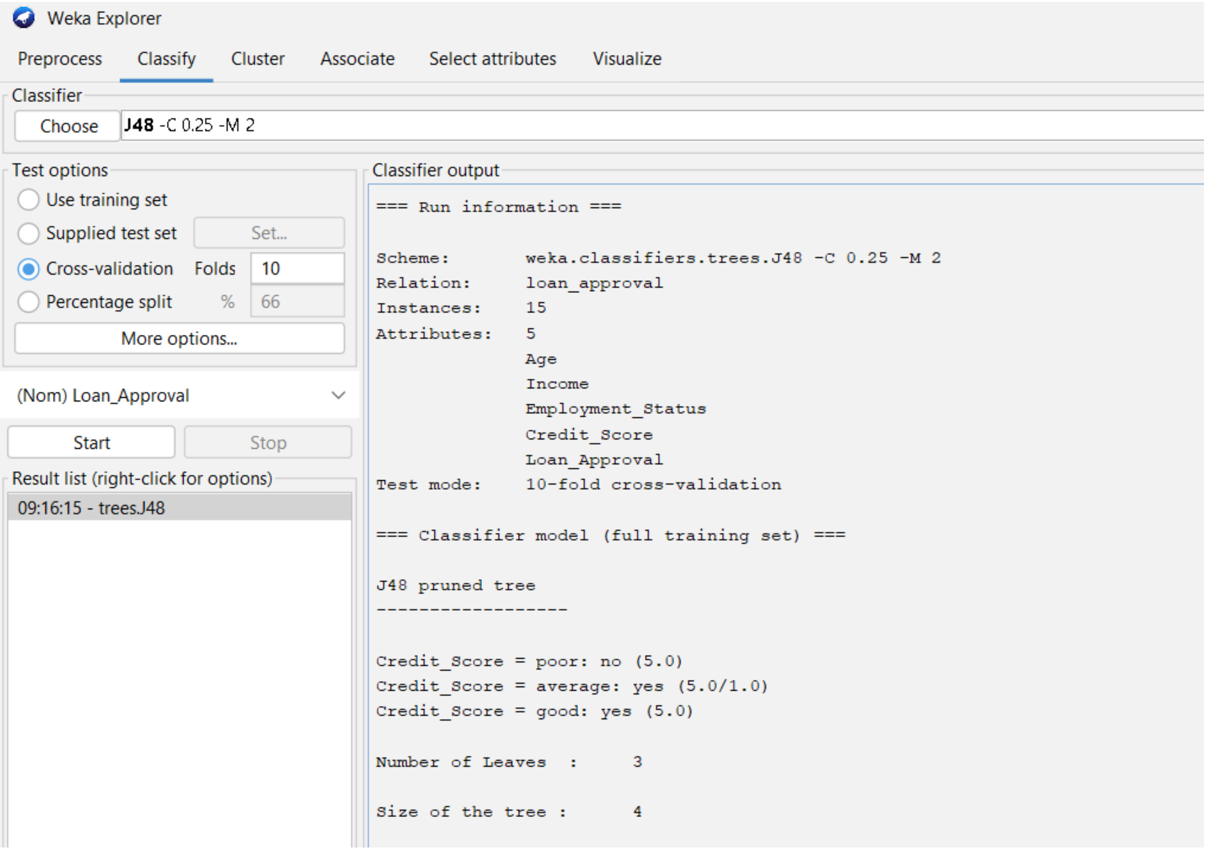
middle,medium,employed,average,yes

middle,high,self-employed,good,yes

old,medium,unemployed,poor,no

young,medium,self-employed,average,yes

middle,low,employed,poor,no



```
Time taken to build model: 0.04 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances          12               80      %
Incorrectly Classified Instances         3               20      %
Kappa statistic                          0.5455
Mean absolute error                      0.2767
Root mean squared error                  0.4667
Relative absolute error                 56.2712 %
Root relative squared error             93.2675 %
Total Number of Instances               15

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.500    0.750      1.000    0.857      0.612    0.611     0.652     yes
                 0.500    0.000    1.000      0.500    0.667      0.612    0.611     0.700     no
Weighted Avg.    0.800    0.300    0.850      0.800    0.781      0.612    0.611     0.671

=== Confusion Matrix ===

 a b   <-- classified as
 9 0 | a = yes
 3 3 | b = no
```

Q4 R Code for Mean, Median, Range, and Boxplot

CODE:

@relation exam_scores

@attribute ClassA numeric

@attribute ClassB numeric

@data

76,51

35,56

47,84

64,60

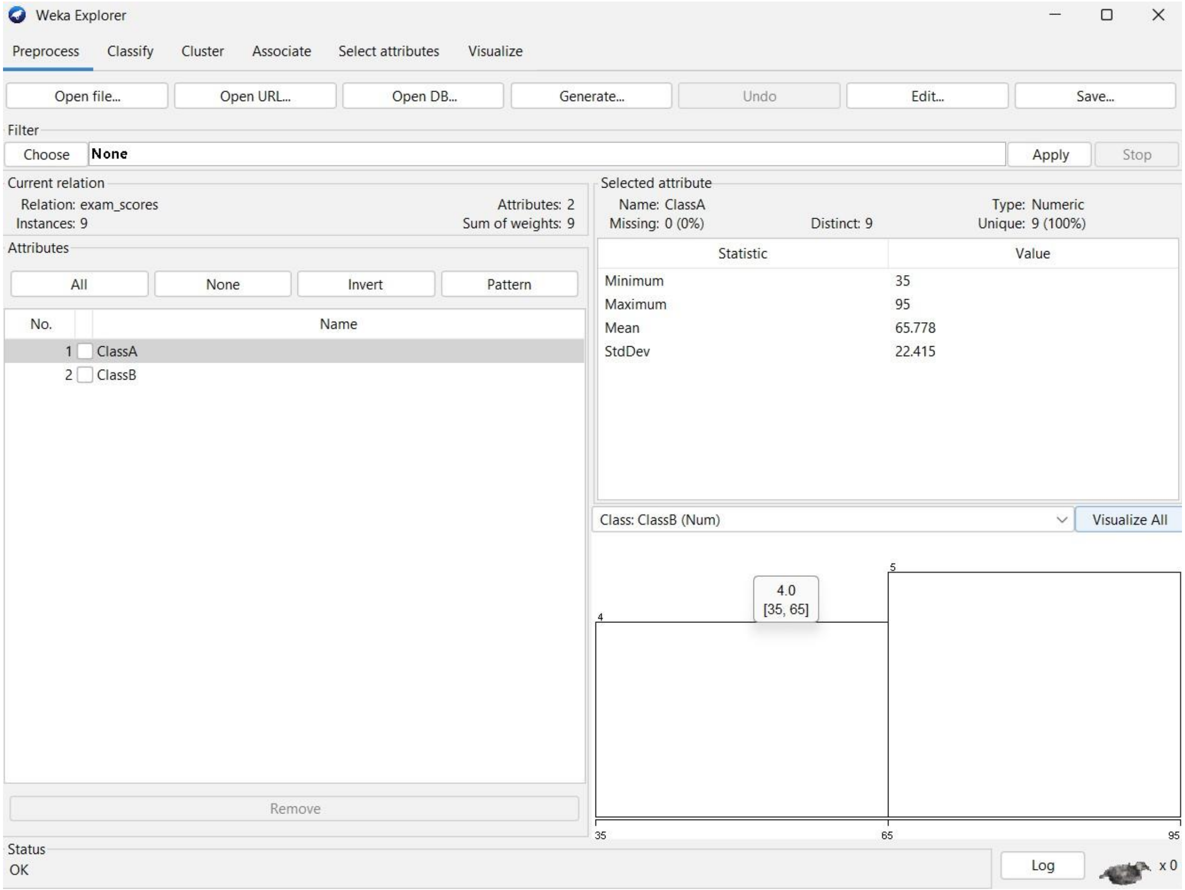
95,59

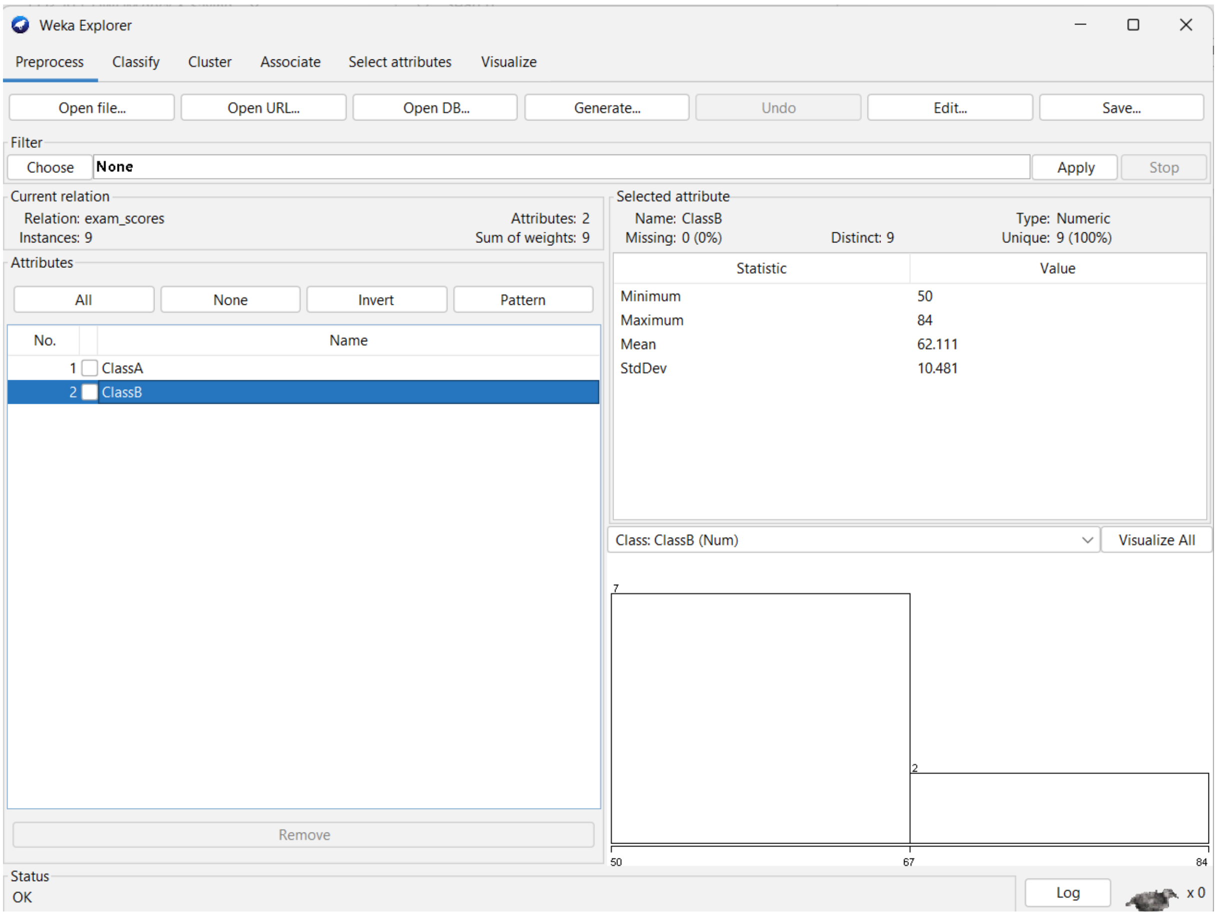
66,70

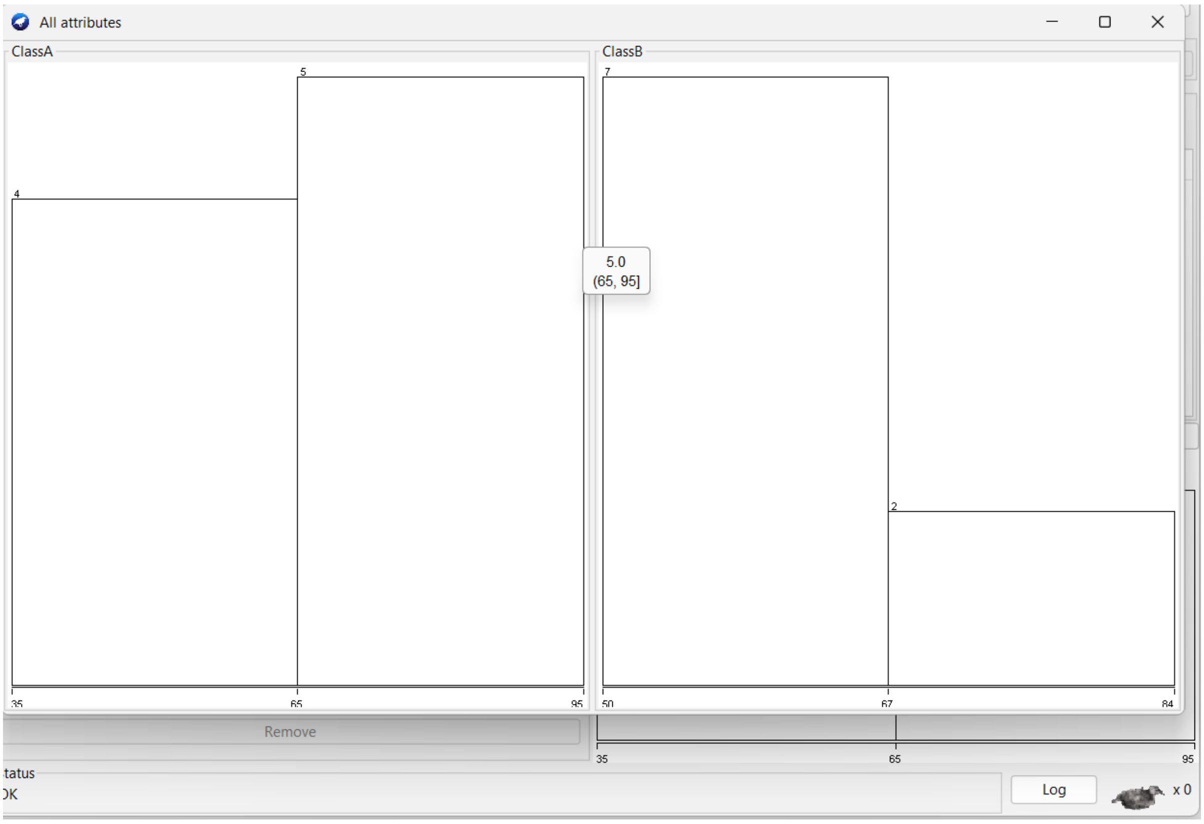
89,63

36,66

84,50


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: exam_scores
Attributes: 2
Instances: 9
Sum of weights: 9
Attributes
All
None
Invert
Pattern
No.
Name
1 ClassA
2 ClassB
Remove
Selected attribute
Name: ClassA
Type: Numeric
Missing: 0 (0%)
Distinct: 9
Unique: 9 (100%)
Statistic
Value
Minimum
35
Maximum
95
Mean
65.778
StdDev
22.415
Class: ClassB (Num)
Visualize All
4.0
[35, 65]
Status
OK
Log
x 0

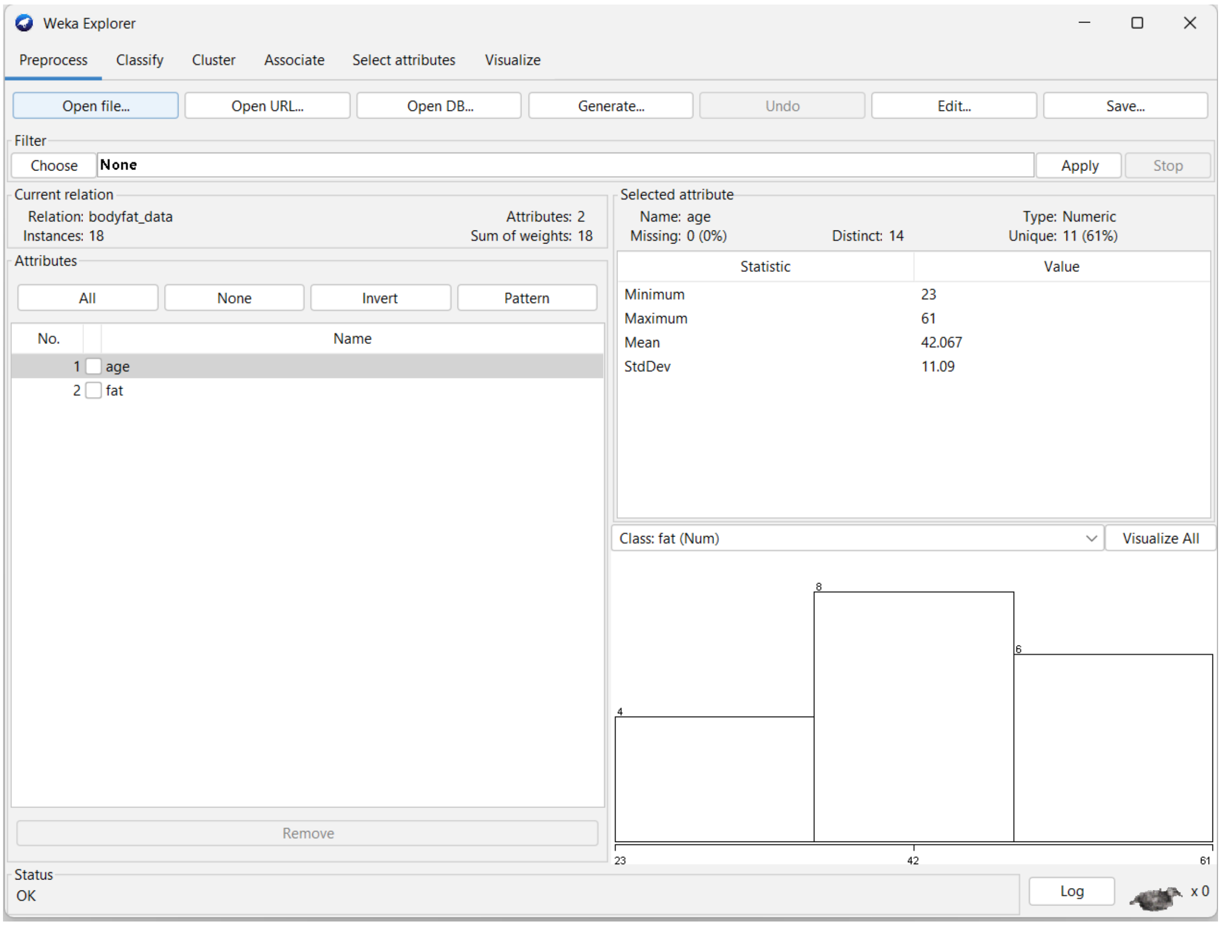

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
None
Apply
Stop
Current relation
Relation: exam_scores
Attributes: 2
Instances: 9
Sum of weights: 9
Attributes
All
None
Invert
Pattern
No.
Name
1 ClassA
2 ClassB
Remove
Selected attribute
Name: ClassB
Type: Numeric
Missing: 0 (0%)
Distinct: 9
Unique: 9 (100%)
Statistic
Value
Minimum 50
Maximum 84
Mean 62.111
StdDev 10.481
Class: ClassB (Num)
Visualize All
7
2
50
67
84
Status
OK
Log
x 0


All attributes
ClassA
5
4
5.0
(65, 95]
35
65
95
ClassB
7
2
50
67
84
Remove
35
65
95
tatus
OK
Log
x 0

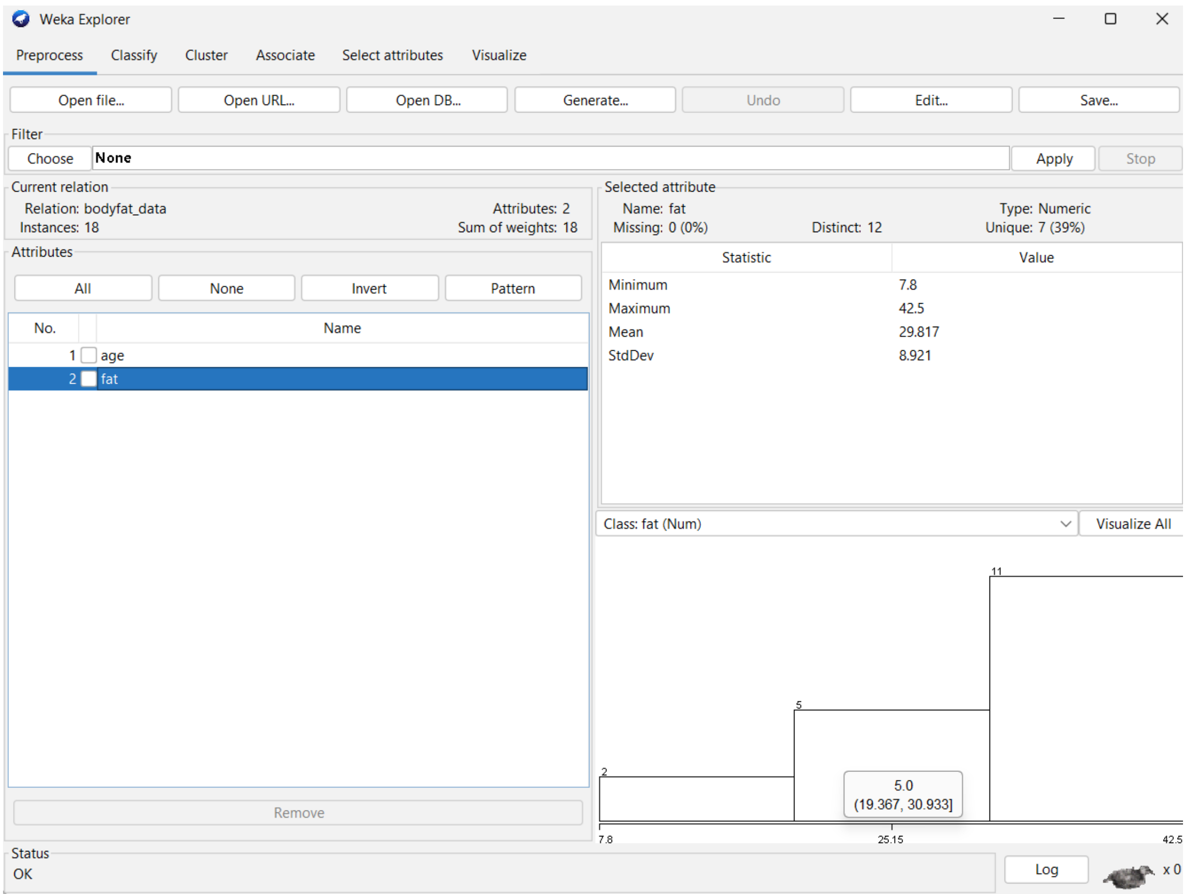
Q5 R Code for Statistical Analysis

CODE:

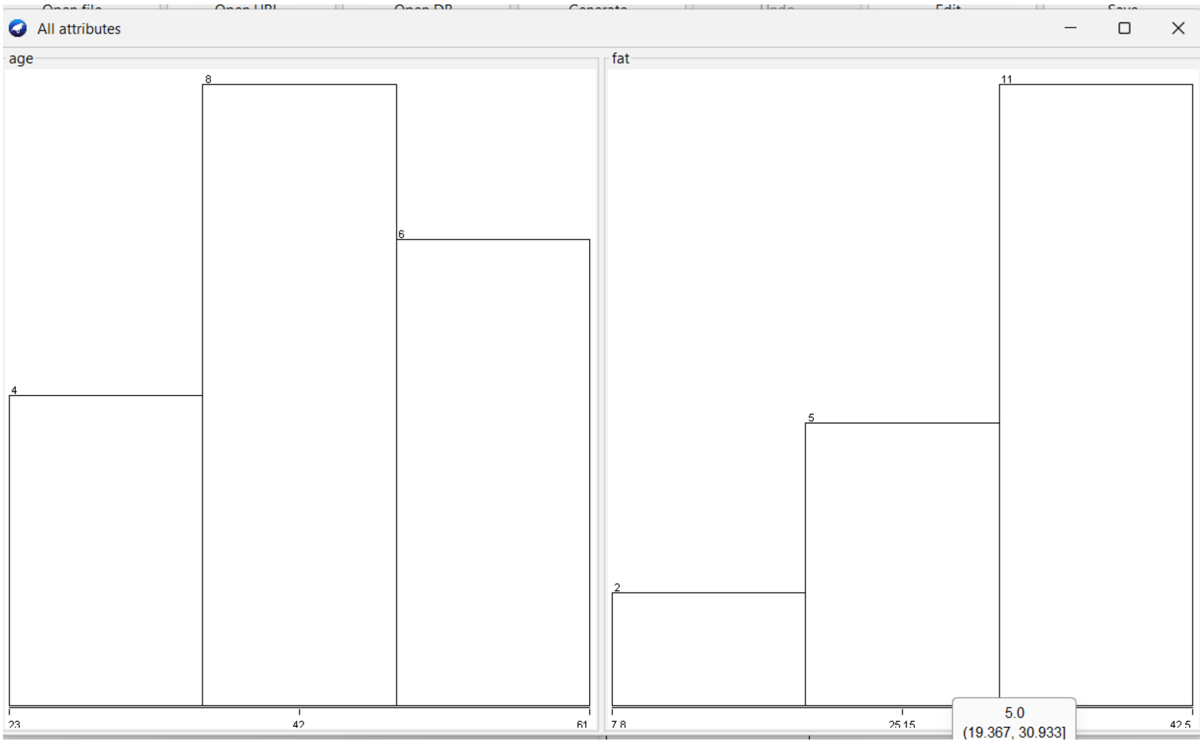
```
@relation bodyfat_data

@attribute age numeric
@attribute fat numeric

@data
23,9.5
23,27.9
27,7.8
37,31.4
41,25.9
47,27.4
49,31.2
50,31.2
34,34.6
42.5,28.8
38,35.2
50,32.8
61,42.5
45,28.8
43,35.2
50,32.8
61,42.5
35.7,31.2
```



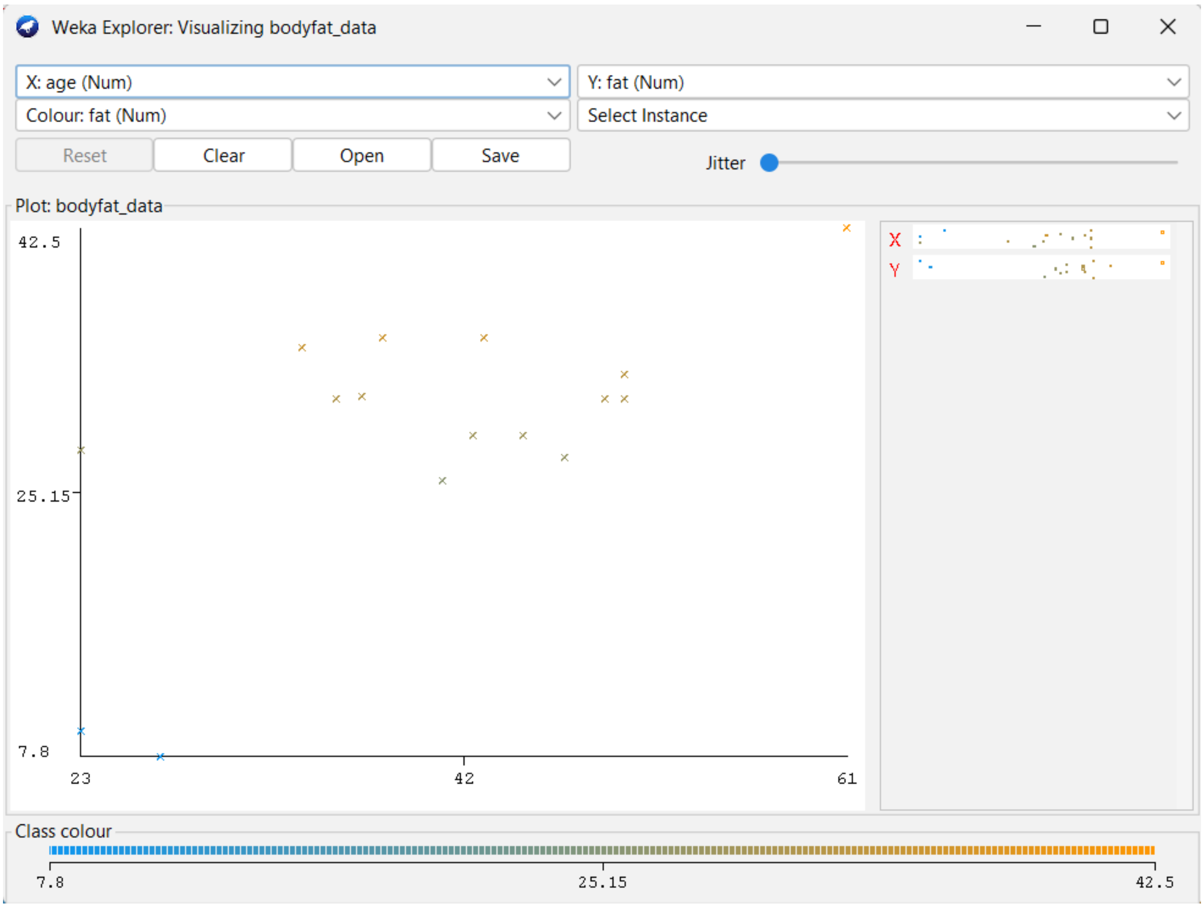
Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Open file... Open URL... Open DB... Generate... Undo Edit... Save...
Filter
Choose None
Apply Stop
Current relation
Relation: bodyfat_data
Attributes: 2
Instances: 18
Sum of weights: 18
Attributes
All None Invert Pattern
No. Name
1 age
2 fat
Remove
Selected attribute
Name: age
Type: Numeric
Missing: 0 (0%)
Distinct: 14
Unique: 11 (61%)
Statistic Value
Minimum 23
Maximum 61
Mean 42.067
StdDev 11.09
Class: fat (Num)
Visualize All
8
6
4
23
42
61
Status
OK
Log
x 0



Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Open file... Open URL... Open DB... Generate... Undo Edit... Save...
Filter
Choose None
Apply Stop
Current relation
Relation: bodyfat_data
Attributes: 2
Instances: 18
Sum of weights: 18
Attributes
All None Invert Pattern
No. Name
1 age
2 fat
Remove
Selected attribute
Name: fat
Type: Numeric
Missing: 0 (0%)
Distinct: 12
Unique: 7 (39%)
Statistic Value
Minimum 7.8
Maximum 42.5
Mean 29.817
StdDev 8.921
Class: fat (Num)
Visualize All
11
5
2
5.0
(19.367, 30.933]
7.8
25.15
42.5
Status
OK
Log
x 0



All attributes
age
fat
8
6
4
11
5
2
23
42
61
7.8
25.15
42.5
5.0
(19.367, 30.933]



Weka Explorer: Visualizing bodyfat_data
X: age (Num)
Y: fat (Num)
Colour: fat (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: bodyfat_data
42.5
25.15
7.8
23
42
61
X
Y
Class colour
7.8
25.15
42.5